

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M.C.A. (CBCS) Degree Examinations

August 2023

22MCA222 – HEALTH CARE ANALYTICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit.

		Unit – I			
		Marks	BT*	CO*	PO*
1.	a) What are the components of healthcare analytics? Explain with examples.	10	1	1	2
	b) List and explain any five imaging modalities.	10	2	1	2
2.	a) Explain the difference between descriptive, diagnostic predictive, and prescriptive analytics in the context of healthcare analytics.	10	1	1	2
	b) Describe some of the applications of healthcare analytics and their importance in healthcare management.	10	2	1	1
		Unit – II			
3.	a) Discuss the concept of mining sensor data in healthcare. What are some of the challenges associated with this process? Explain.	10	2	2	1
	b) List and explain the types of Biomedical Signals.	10	2	2	1
4.	a) List and explain the different connectivity methods.	10	2	2	2
	b) Explain with a neat diagram the typical processing pipeline for rs-fMRI processing.	10	2	2	2
		Unit – III			
5.	a) What is genomic data? List and explain the different ways to generate genomic data.	10	2	3	2
	b) What is Clinicogenomic Integration? Discuss the issues and challenges in Integrating Clinical and Genomic data.	10	2	3	1
6.	a) Explain the different stages of data integration, highlighting the advantages and their disadvantages.	10	2	3	2
	b) Explain the Stage of Dimensionality reduction. Also highlight its advantages and disadvantages.	10	2	3	1
		Unit – IV			
7.	a) Explain the concept of natural language processing for health care with its general workflow and its components.	10	2	4	2
	b) List and explain the components of EHR.	10	2	4	1
8.	a) Discuss the challenges of processing clinical reports. Also list the clinical applications.	10	2	4	2
	b) Discuss the role of social media for the analysis, detection and tracking of infectious disease outbreaks.	10	2	4	1
		Unit – V			
9.	a) Explain the various statistical prediction models used in healthcare data analytics.	10	2	5	2
	b) Discuss the importance of visual analytics in healthcare and its applications in clinical workflow, clinicians, and patients.	10	2	5	2
10.	a) Describe the popular temporal pattern mining methods.	10	2	5	2
	b) Explain the alternative prediction models given below with reference to data analytics for healthcare.				
	i) Decision trees				
	ii) Artificial neural networks				
	iii) Cost-sensitive learning.	10	2	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
